

Efficiency and acceleration in the accumulation of children’s early vocabulary

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Abstract

Children learn hundreds of words over the first few years of their life. This process begins slowly but soon children are acquiring multiple words per day. Yet there are also wide variations between children, and the nature of this variation has not been accurately characterized. We present a statistical model describing this process as accelerating accumulation, with children varying in their overall learning efficiency and their acceleration. This model provides a good characterization of longitudinal growth across three languages and gives insight into the sources of sociodemographic variation. Using dense observations of children’s early language environment and word recognition speed, the model also can be used to estimate how much variation between children is due to differences in language input and processing. In contrast to children, large language models are not well-described by this model: they show standard accumulator dynamics with no acceleration and very limited variability. In sum, our model provides a quantitative characterization of key signatures of early vocabulary growth, providing a target for future mechanistic learning models.

Children’s early vocabulary grows very rapidly during early childhood, with large variation between children (Fenson et al. 1994, 2007; Frank et al. 2021). The growth has theoretical, practical, and methodological importance. From a theoretical perspective, growth mechanisms shed light on children’s fundamental learning mechanisms and how they change across childhood (McMurray 2007; Bloom 2002; Frank et al. 2021). From a practical perspective, early vocabulary is an important precursor – and predictor – of academic achievement [Marchman and Fernald (2008);CITES]. From a methodological perspective, vocabulary also affords a unique psychometric opportunity to study human variation because, unlike nearly all important human psychological metrics, there is a true zero point – a time before which children know no words. Thus, absolute levels of variability can be calculated.

One dominant viewpoint is that children’s vocabulary growth can be characterized as a process of accumulation (McMurray 2007; Mitchell and McMurray 2009; Hidaka 2013; Mollica and Piantadosi 2017; Kachergis, Marchman, and Frank 2022). Each distinct word (type) can be conceptualized as a “bucket,” and every token of language input then is a “drop” that falls into its respective bucket. When each bucket is full, the word is learned (McMurray 2007; Kachergis, Marchman, and Frank 2022). These models have been used to recover interpretable differences in “bucket size,” often based on word properties including concreteness and syntactic category (Hidaka 2013; Kachergis, Marchman, and Frank 2022). They also provide a graded explanation for the widely-observed “vocabulary spurt” phenomenon – in which children’s rate of vocabulary learning appears to increase dramatically (Frank et al. 2021; Ganger and Brent 2004; Gómez Díaz et al. 2024).

The accumulator view connects with a viewpoint in which the quantity of language that children

hear plays a strong causal role in the speed of their learning (Fernald and Marchman 2012). Indeed, there is systematic evidence that variation in children’s input is related to variation in vocabulary [Fernald and Marchman (2012); Weisleder and Fernald (2013); CITES]. Studies of input have identified systematic variation in *quantity* in terms of sheer amount of tokens as well as variation in *quality*. Measuring what makes input high quality has been challenging, but higher quality input likely has higher lexical diversity, longer sentences, more declarative and interrogative content relative to imperatives, and more elaborated discourse [LOTS OF CITES].

A naïve interpretation of the accumulator view would suggest that most of the variance in vocabulary growth should be related to input. Two recent meta-analyses complicate this view, however. Anderson et al. (2021) showed that only a modest proportion of the variance in vocabulary outcomes is related to input quantity (XYZ) and quality (XYZ). A newer meta-analysis of quantity specifically confirms and extends the proportion of variance due to quantity to 4–7% of total variation in vocabulary (Coffey and Snedeker 2026). Thus – as we might have supposed based on other research on human variation – not all variation in vocabulary between children can be attributed to input differences.

As this discussion indicates, despite the promise of the accumulator viewpoint, it has not been used successfully to understand variation between individuals. Instead, the majority of applications have focused on variation between words. Further, the rapidity of children’s early word learning stands as a rebuke to this view. Sometimes a small number of examples is enough to form a lasting representation (Markson and Bloom 1997; Woodward, Markman, and Fitzsimmons 1994). Perhaps only a small number of high-quality instances is sufficient for learning (Trueswell et al. 2013; Mollica and Piantadosi 2017), but if so, why do some words take much longer to learn?

The goal of this work is to address these puzzles through systematic investigation and extension of accumulator models. All models seek to explain the observation that the overall size of children’s vocabulary increases supra-linearly. Does this growth imply developmental changes in the underlying process?

Following previous work, we start with the idea that children’s *internal learning ability* (not just their observed vocabulary growth) accelerates across development (Hidaka 2013; Mayor and Plunkett 2010). In the bucket metaphor, the drops get bigger as children get older. We begin by extending the psychometric accumulator model of Kachergis, Marchman, and Frank (2022): our key addition is to describe variation between individual children by modeling both their individual learning efficiency and the acceleration of their learning. This extension defines a longitudinal growth model that can be fit to data, and that provides a simple reconciliation between slow accumulation and fast mapping: the number of examples of a word necessary for learning declines by multiple orders of magnitude from age one to age three.

To constrain our model, we use data from the MacArthur-Bates Communicative Development Inventories [MB-CDI; Fenson et al. (1994); Marchman, Dale, and Fenson (2021)], a popular parent report instrument in which parents are asked to indicate which words their child produces and understands. Here we focus exclusively on production, which is used with older children and which tends to be a more reliable measure (Frank et al. 2021). We make use of longitudinal data from Wordbank (Frank et al. 2017), an open repository of data from the MB-CDI.

To address questions about the relationship between language input and growth, we fit a Bayesian

version of our growth model. We make use of new data resources to estimate the magnitude of linkages between input and vocabulary growth. First, we estimate plausible population values based on independent estimates of population variation in language input (Coffey et al. 2024). In addition, we use measures of individual children’s language input from egocentric videos (Long et al. 2024) and day-long home recordings (Egan-Dailey and Bergelson 2025; Adams et al. 2018; Fernald, Marchman, and Weisleder 2013) to estimate the input-uptake relationship more precisely for a subsample of children.

Overall, our analysis supports a view in which children’s vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. We end by comparing the fit of our models on human data to their fit on large-language models. Making use of GPT-2 models trained on human developmental data, we show that – in contrast to children – these models do appear to function as pure accumulators. This analysis exposes a critical disanalogy between LLMs and human learners and suggests a potential route for understanding the vast “data gap” between children and large language models (Frank 2023; Warstadt et al. 2023).

Model

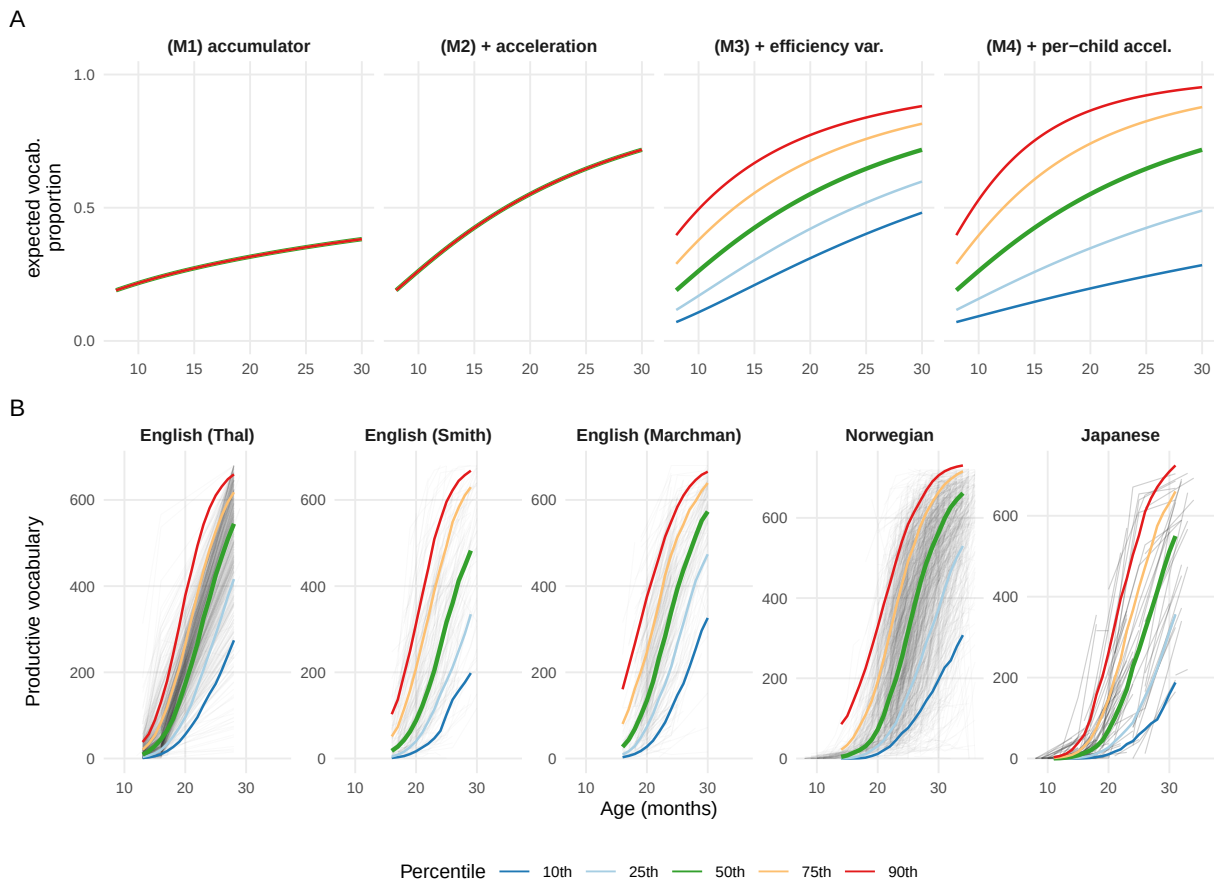


Figure 1

We present a nested set of accumulator-type models, many of which have been presented in the literature. As we show below, each refinement of the model contributes to its ability to describe the variability between children. We start with the Rasch model from psychometrics (Rasch 1960; Bond and Fox 2013), which assumes that each child i has a learning ability θ_i and each word j has a difficulty δ_j . The probability that child i produces word j is

$$P(w_{i,j} = 1 \mid \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$

Following Kachergis, Marchman, and Frank (2022), we can write our first model as a Rasch-style accumulator model (Model 1), specifying that θ_i is not a single parameter but instead a function of accumulation over time. In the simplest form:

$$\text{M1: } \theta_i(t) = \log(r_i) + \log(t/a_0)$$

where r_i is the child’s rate of language input (in tokens per month) and t is the child’s age (in months, anchored at a_0); note that we use log age to simulate linear accumulation because $\exp(\theta) = r \cdot t$. This form instantiates the basic hypothesis that the only thing that changes over time is more input, and the only source of variation between children is variation in rate.

We compare this model to an accelerating accumulator model (Model 2), in which the amount learned from each exposure increases with time by an exponent κ .

$$\text{M2: } \theta_i(t) = \log(r_i) + \kappa \log(t/a_0)$$

When $\kappa > 1$, each successive month yields more accumulation than the last.

Neither of these models has any way to capture variation across children beyond variation in input rate. Thus, we augment the basic accelerating accumulator, rewriting it as the sum of either one or two individually variable quantities:

$$\text{M3: } \theta_i(t) = \xi_i + \kappa \log(t/a_0)$$

$$\text{M4: } \theta_i(t) = \xi_i + \kappa_i \log(t/a_0)$$

The intercept term in both models $\xi_i = \log(r_i) + \log(\alpha_i)$ can then decompose individual variation into both input rate-related and other variation in children’s efficiency. Similarly, in M4, the slope term $\kappa_i = \kappa + \gamma \log r_i + \zeta_i$ adds *individual variation in acceleration*, decomposing acceleration into a population term, input-related acceleration, and a child-specific residual acceleration ζ_i (the part not explained by input). The coefficient γ then captures the strength of the relationship between input and acceleration.

Model 4 instantiates the idea that children vary in their base rate of efficiency and their acceleration, creating variant of a standard longitudinal growth model in which children vary on their slope and intercept (Grimm, Ram, and Estabrook 2016). Note that, while we define the relationship of ξ_i and

κ_i to a child’s input r_i , we can nevertheless fit the simplest form of the model by just allowing these parameters to vary as random child-level slopes and intercepts.

As we will show, this model provides a good description of variation in children’s longitudinal growth trajectories. Figure 1 (top) shows schematic predictions from a pure accumulator model (M1), an accelerating accumulator (M2) in comparison to models including child-level variation in efficiency (M3) and both efficiency and acceleration (M4). Each model’s predictions are represented as ability scores (θ) across age and then also projected into proportion correct scores on a discrete set of vocabulary items, representing an observed accuracy score that is subject to compression at floor and ceiling.

A variety of previous work has focused on estimating the properties of individual words (Goodman, Dale, and Li 2008; Hidaka 2013; Roy et al. 2015; Braginsky et al. 2019; Kachergis, Marchman, and Frank 2022; Kachergis et al. 2022). These analyses reveal that there is predictable word-level difficulty information but that only a modest portion of the variation between words is carried by frequency information. Words can be hard for a child to acquire because they vary also in the complexity and concreteness of their meanings, their syntactic category and syntactic context, their phonology, and many other factors. In the current work, we estimate word difficulty as a free parameter, and focus instead on variation between children.

Our analysis generally depends on the availability of longitudinal data at the item level. The different models described above are not distinguishable from the cross-sectional, aggregate data that are typically considered in many analyses of early vocabulary. First, aggregate data do not allow for the estimation of individual word difficulties, thus the distribution of difficulties remains latent and different difficulty distributions can be posited to explain the accelerating shape of vocabulary growth (McMurray 2007; Mitchell and McMurray 2009). Second, cross-sectional data do not allow for identification of whether individual children’s variation is due to differences in efficiency (intercept) or acceleration (slope) (see Supplemental Figure 1).

Results

Children’s longitudinal growth reflects differences in both efficiency and acceleration

We first retrieved longitudinal data on monolingual, typically-developing children’s vocabulary from Wordbank (Frank et al. 2017), an open archive of data from the MacArthur-Bates Communicative Development Inventory and its international adaptations. Table 1 shows the four longitudinal datasets in Wordbank that contained more than > 100 children with multiple observations.

Table 1: Characteristics of all datasets used in the paper. N and ages reflect the analysis samples (post-exclusion). Cross-sectional rows pool all contributed Wordbank datasets for a language, subsampled to at most 1,200 children. For AM2018 and FMW2013, which enter both the input and processing analyses with slightly different inclusion criteria, N is the union of the two subsamples.

Citation	Language	N	Admins	Age (months)			Long.	Input	Recordings	LWL sessions
				Mean	Min	Max				
Wordbank longitudinal										
Thal et al. (20XX)	English (American)	653	1,947	19.1	12	29	yes			
Smith et al. (20XX)	English (American)	314	878	21.9	16	30	yes			
Marchman et al. (20XX)	English (American)	314	714	22.6	16	30	yes			
Simonsen et al. (2014)	Norwegian	1,676	6,347	25.1	8	36	yes			
Hagihara et al. (2023)	Japanese	178	425	17.4	8	34	yes			
Input / processing										
Long et al. (2024)	English (American)	22	103	18.6	9	32	yes	headcam	5,739	
Egan-Dailey & Bergelson (2025)	English (American)	44	514	12.2	6	18	yes	LENA	525	
Adams et al. (2018)	English (American)	67	167	20.1	13	27	yes	LENA	126	440
Fernald et al. (2013)	English (American)	60	142	23.2	18	30	yes	LENA	51	93
Fernald & Marchman (2012)	English (American)	119	393	22.9	14	30	yes			419
Wordbank cross-sectional										
Frank et al. (2017)	Arabic (Saudi)	258	258	11.9	8	16				
Frank et al. (2017)	Cantonese	1,200	1,200	23.4	16	30				
Frank et al. (2017)	Catalan	1,200	1,200	19.6	8	30				
Frank et al. (2017)	Croatian	627	627	19.0	8	30				
Frank et al. (2017)	Czech	493	493	23.0	16	30				
Frank et al. (2017)	Danish	1,200	1,200	20.9	8	36				
Frank et al. (2017)	Dutch	446	446	15.6	8	31				
Frank et al. (2017)	English (American)	1,200	1,200	19.3	8	30				
Frank et al. (2017)	English (Australian)	1,200	1,200	20.5	12	30				
Frank et al. (2017)	Estonian	1,200	1,200	23.3	16	30				
Frank et al. (2017)	Finnish	234	234	16.5	9	24				
Frank et al. (2017)	French (French)	1,200	1,200	21.6	8	30				
Frank et al. (2017)	German	1,181	1,181	24.1	18	30				
Frank et al. (2017)	Hebrew	534	534	27.4	11	35				
Frank et al. (2017)	Hungarian	369	369	23.1	16	30				
Frank et al. (2017)	Italian	1,200	1,200	20.5	7	36				
Frank et al. (2017)	Japanese	846	846	20.4	8	36				
Frank et al. (2017)	Kigirيامa	205	205	16.1	8	30				
Frank et al. (2017)	Korean	1,200	1,200	23.0	8	36				
Frank et al. (2017)	Latvian	683	683	22.3	8	36				
Frank et al. (2017)	Mandarin (Beijing)	1,056	1,056	23.0	16	30				
Frank et al. (2017)	Mandarin (Taiwanese)	1,200	1,200	21.8	8	36				
Frank et al. (2017)	Norwegian	1,200	1,200	23.3	8	36				

(continued)

Citation	Language	N	Admins	Age (months)			Long.	Input	Recordings	LWL sessions
				Mean	Min	Max				
Frank et al. (2017)	Portuguese (European)	1,200	1,200	19.4	8	30				
Frank et al. (2017)	Russian	1,200	1,200	21.4	8	36				
Frank et al. (2017)	Slovak	1,200	1,200	20.0	8	36				
Frank et al. (2017)	Spanish (Argentinian)	783	783	23.3	16	33				
Frank et al. (2017)	Spanish (European)	1,005	1,005	18.4	8	30				
Frank et al. (2017)	Spanish (Mexican)	1,200	1,200	18.9	8	30				
Frank et al. (2017)	Swedish	1,200	1,200	18.9	8	28				
Frank et al. (2017)	Turkish	1,200	1,200	21.4	8	36				

Taking advantage of the equivalence of Rasch models to generalized linear mixed effects models (De Boeck et al. 2011), we fit the set of models shown in Figure 1 as a series of logistic regressions. As described above, this initial set of fits do not attempt to estimate the relationship with input but instead simply attempt to describe the distribution of vocabulary growth curves. Each model predicted the probability of an individual child producing a particular word as a function of age (either unmodified in M1 or multiplied by a constant in M2) and a random effect of word difficulty. Successive models add random intercepts (M3) and slopes (M4) by child.

Model 4, which incorporated variation in acceleration and efficiency, was the best fitting model across all datasets, with AIC and BIC differences shown in Table 2. Figure 1 (bottom) shows the fit of the four model variants; the increasing spread of children’s trajectories with age is especially noticeable for low-vocabulary older children [who may be classified as “late talkers”; Fernald and Marchman (2012); Rescorla (2011)].¹

Table 2

Model	Age	ΔAIC				
		Thal	Smith	Marchman	Norwegian	Japanese
M1. accumulator	—	703,220	273,732	251,211	2,281,606	94,503
M2. + acceleration	linear	206,553	179,872	134,596	1,377,615	41,169
M2. + acceleration	log	204,930	178,471	133,877	1,348,013	39,054
M3. + efficiency var.	linear	35,856	17,418	13,752	130,625	3,850
M3. + efficiency var.	log	34,219	15,356	13,135	115,116	3,227
M4. + acceleration var.	linear	1,134	553	0	4,301	169
M4. + acceleration var.	log	0	0	30	0	0

Efficiency and acceleration relate to separate demographic predictors

Children’s vocabulary is known to be related to a number of sociodemographic predictors, including sex and maternal education (which is interpreted as a proxy for socioeconomic status). In particular, female children show consistently larger vocabularies than male children across dozens of languages and countries (Eriksson et al. 2012; Frank et al. 2021). Similarly, children of more educated mothers tend to show larger vocabularies, though this effect is somewhat more heterogeneous across countries and languages (Frank et al. 2021).

Our longitudinal model offers a summary of individual children’s growth in terms of both their efficiency (intercept) and their acceleration (slope). This decomposition allows us to ask whether these two demographics effects – sex and maternal education – are seen in efficiency, acceleration, or both. For each child in our analysis, we extracted the best linear unbiased prediction (BLUP) from our models. We then regressed intercept and slope terms against each demographic predictor separately for each language. Figure 2 shows the coefficients from this analysis, along with a cross-language meta-analytic estimate.

¹Following the literature, we primarily investigate linear accumulator models, which show power-law growth of ability over time because θ is a function of $\log(t)$. However, it is also possible that ability grows linearly in time, which would lead to exponential increases in the probability of word production. We fit linear-time models, but found that they were dispreferred statistically, though differences were relatively small within the restricted age range for which we have data (δAIC 0–4,301; see Supplemental Figure 2 for model fits).

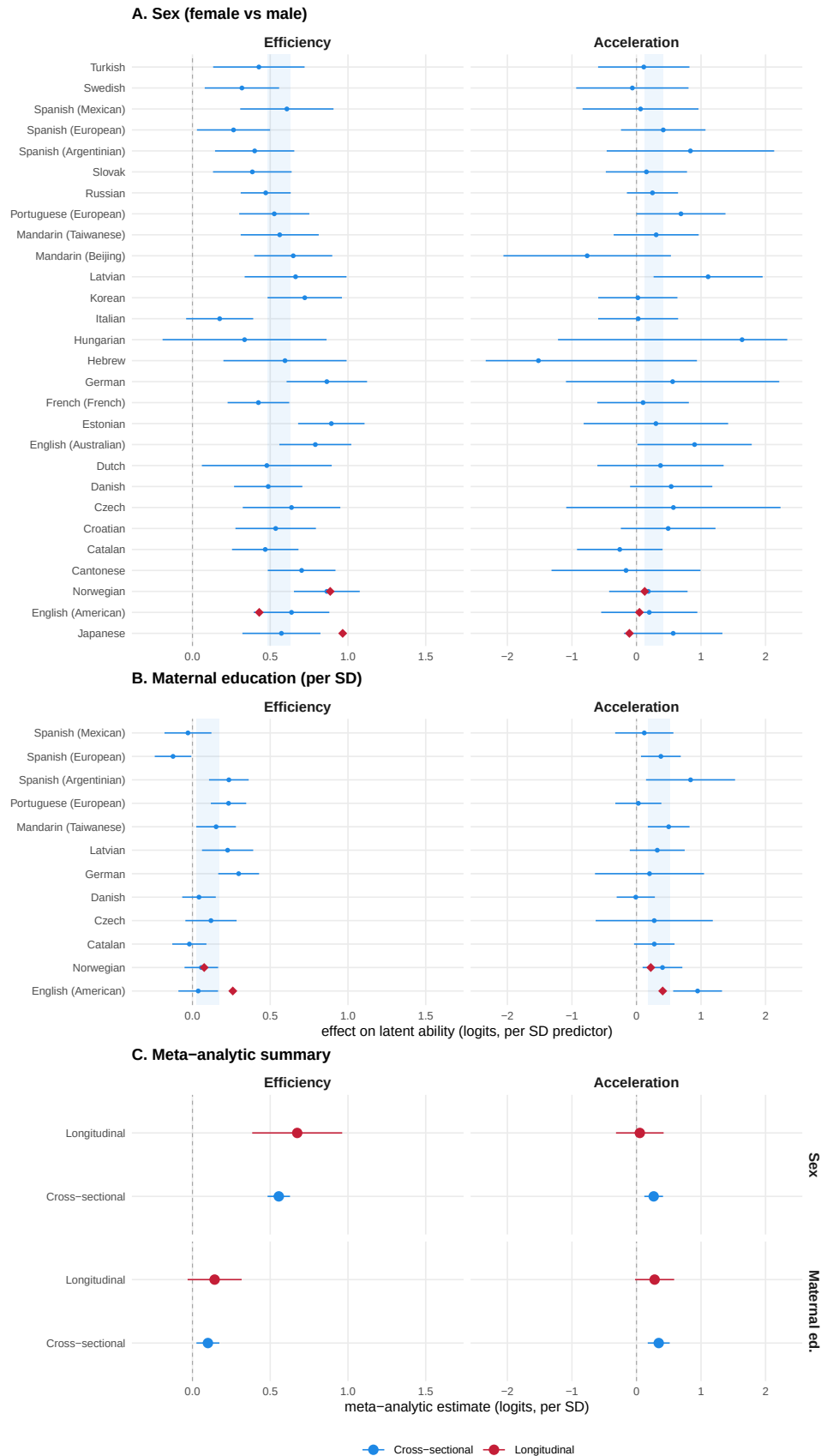


Figure 2: Regression coefficients predicting individual children's efficiency and acceleration as a function of gender and maternal education. Black dots show meta-analytic estimates from a random-effects meta analysis. Error bars show 95% confidence intervals.

Sex effects show the predicted female advantage across all four datasets, but this effect manifests itself only in efficiency ($\beta = -0.33$, $p = < .001$) not acceleration ($\beta = 0$, $p = .939$). In contrast, though maternal education data were only available for two languages, effects were numerically present in both measures, though neither was significant (efficiency: $\beta = 0.06$, $p = .157$; acceleration: $\beta = 0.03$, $p = .096$). One interpretation of these findings is that sex effects reflect a small constant offset in average ability. In contrast, socioeconomic differences as reflected in maternal education might continue to compound across development (Fernald, Marchman, and Weisleder 2013; Hart and Risley 2003).

Population input-related variation corresponds to meta-analytic estimates

In our prior model fits, we fit population variation in efficiency and acceleration without considering the connection to language input. We next consider whether this relationship can be estimated at a population level. We use markov chain monte carlo to fit the full version of model 4, with both efficiency and acceleration linked to input variation (see Supplemental Methods for full model specification). We estimate these linkages by assuming a known level of population variation in language input rate: σ_r . We can then examine the variance due to input variation vs. population by fitting the parameters $\pi_\xi = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2}$ and $\pi_\kappa = \frac{\sigma_\zeta^2}{\sigma_\zeta^2 + \sigma_r^2}$, which respectively define the proportion of population variation in efficiency and acceleration due to input variation.

Estimating children’s input range is notoriously tricky, since the rate of language that children hear varies across cultures, families, activities, and measurement occasions (Coffey et al. 2024). To establish a range of plausible values for σ_r , we surveyed the literature for plausible estimates [CITES;(sperry2018?);Coffey et al. (2024)]. In our main simulations we set $\sigma_r = .53$, a value drawn from (sperry2018?), who estimated a mean of 419k tokens per month (± 1 SD: 329k – 957k). See Supplemental Methods for other input estimates.

[results](#)

[Sensitivity checks]

[relation to meta-analysis]

Input modestly predicts efficiency and acceleration for individual children

While the population estimates of input-sensitivity given above match our expectations from meta-analyses, they do not offer a direct estimate of input couplings within individual children. For this, we turn to a set of four longitudinal datasets that include dense home recordings (see (tab-datasets?)). Three of these include day-long audio recordings using LENA (Language Environment Analysis) recorders, which offer an automated adult word count estimate that has been validated by a number of transcription studies (cristia2021?). The last includes short video recordings of egocentric video recorded in children’s homes, with word counts estimated from automated transcripts. Each of these provides a noisy proxy for total home input, but nevertheless allows us to estimate models that include individual input.

We fit the same model linking input estimates to variation in efficiency and acceleration, but this time with observations of r_i for each child. Results for $\pi_{\alpha}lpha$ and π_zeta are shown in Figure 3.

RESULTS

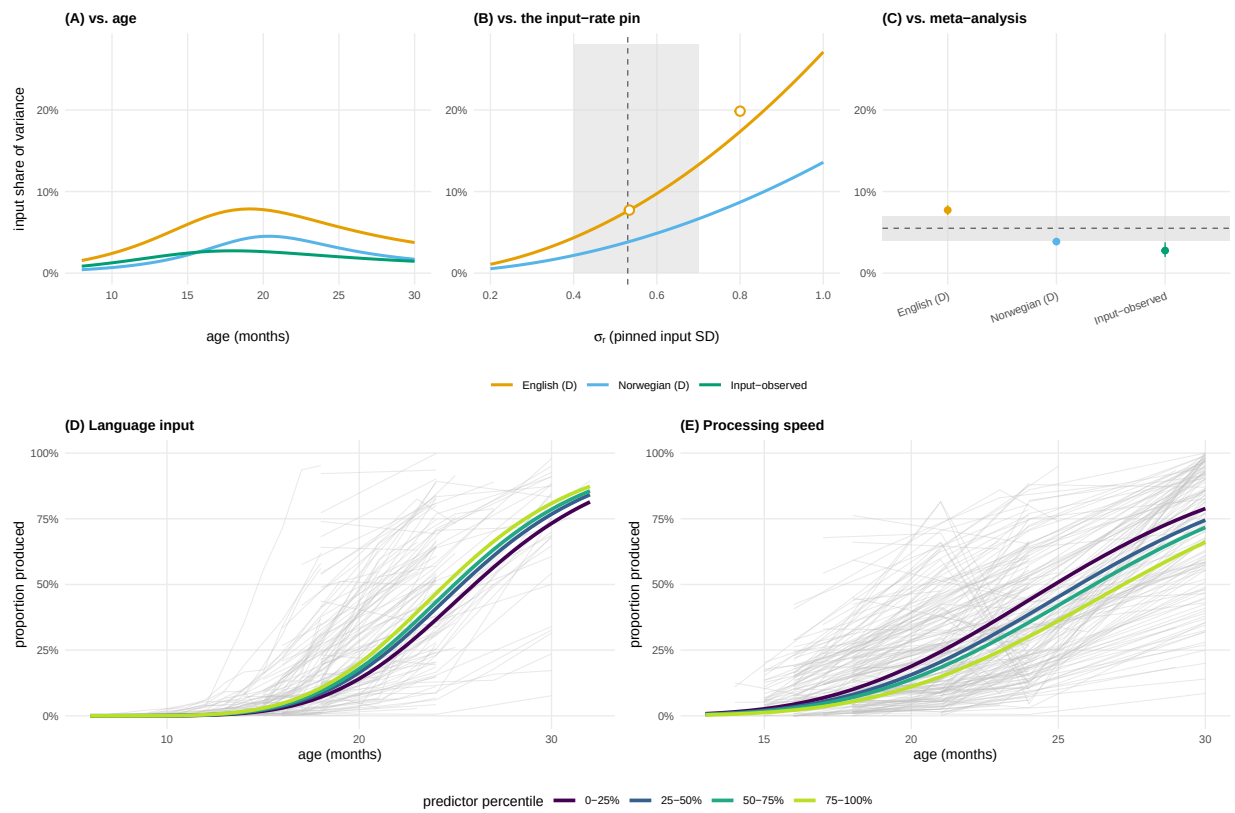


Figure 3

Processing speed is strongly associated with efficiency

Two of the datasets we included also include longitudinal reaction time data from the looking-while-listening paradigm (LWL), archived in the Peekbank repository (zettersten2022?). In LWL, children see two pictures (e.g. a book and a car) and their eye movements are recorded as they hear a sentence that refers to one of the pictures (e.g., “look at the book”). If the child is looking at the distractor image (the car), then their reaction time to shift their eyes to the target provides an index of their speed of word recognition (fernauld2008?; frank2026?).

We fit a modification of our model to this subset of the data (N=XYZ). Efficiency and acceleration terms included *both* input and LWL reaction time as predictors.

Growth in ability leads to rapid changes in learning times for individual words

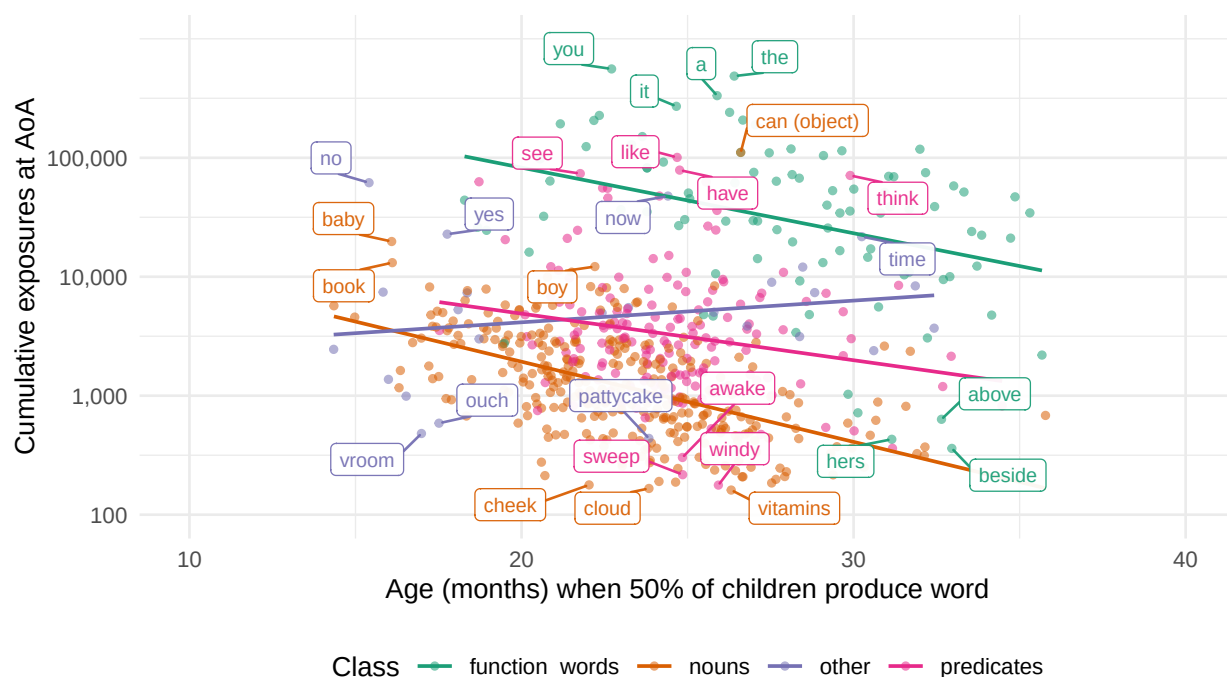


Figure 4

As an illustration of the changes in efficiency and acceleration that we observe in our data, we computed the average proportion of exposures necessary for a child to learn a word across ages. Figure 4 shows a model-derived estimate of the average number of times a word is heard before more than 50% of children produce it (See SI: Changes in Word Efficiency), using frequency estimates from the CHILDES Corpus (MacWhinney 2000; Sanchez et al. 2019).

Across different word classes (e.g., nouns, verbs, closed class words), efficiency increases exponentially, such that early-learned words such as “book” may be heard tens of thousands of times before they are produced, while later learned words such as “vitamin” might only be heard hundreds times before being produced. Indeed, this analysis likely understates the degree of change in efficiency as it assumes that each exposure to a word is equally meaningful. In fact, individual exposures vary widely in how much information they give. Sometimes words are overheard rather than directed to the

child, or lack physical context to allow for clear inference (**effective_learning_instances_erika?**; **truewell2014?**; Mollica and Piantadosi 2017). Thus, in line with both intuition and experiments, the total number of exposures required on average to learn a word might indeed be only a small handful by age three (Markson and Bloom 1997; Bloom 2002).

Large language models do not show acceleration in their vocabulary growth

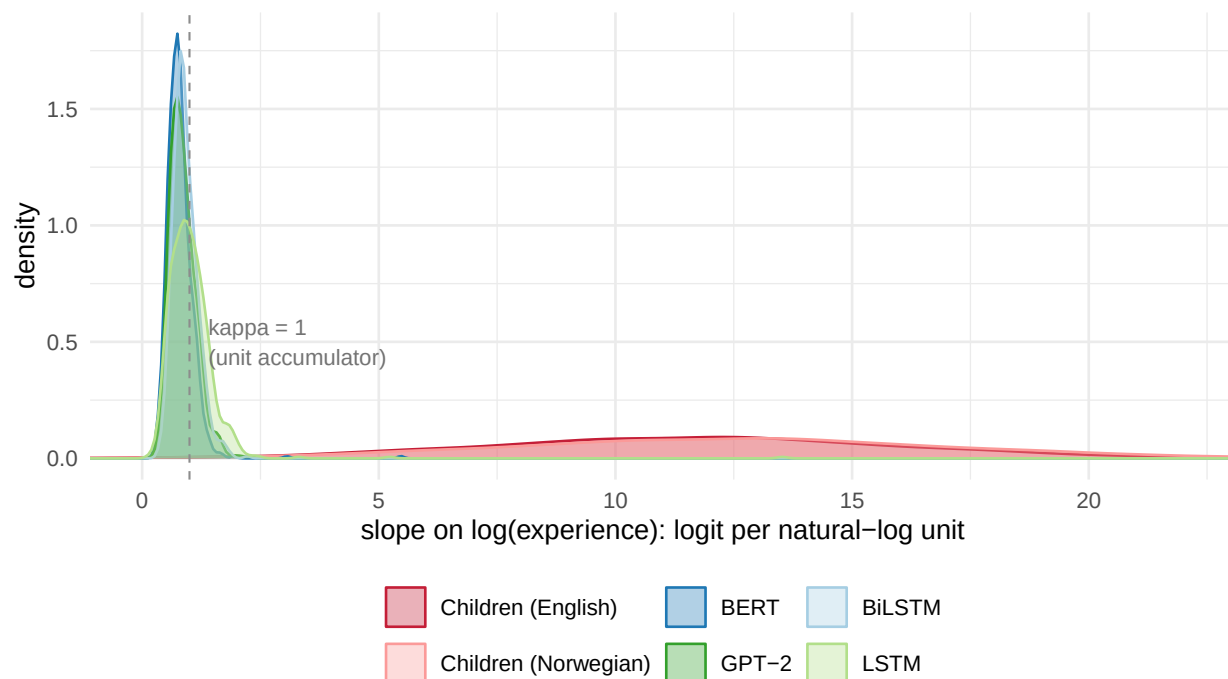


Figure 5: Comparison of acceleration slopes (κ) estimated in our experiments for children in English and Norwegian, vs. LLM slopes per model estimated by Chang & Bergen (2022) and in our experiments.

Small-scale language models trained on developmentally-realistic corpora have become a useful tool for understanding early language development (**huebner2022?**; **futtrell2026?**; **frank2025?**). For our final experiments, we were interested in whether large language models show the same patterns of developmental change in efficiency or acceleration. We made use of work by Chang and Bergen (2022), who extracted per-word acquisition curves from language models. They computed logistic fits to the surprisal (negative log probability) of individual words across training, estimating

We can compare

(**evanston2023?**) investigated

We end by providing a quantitative comparison between within-child human learning and LLM scaling laws, showing that children’s early language learning is strikingly different from LLM scaling: it shows both steeper scaling and greater variance in growth rate.

Discussion

What we did

Longitudinal growth models efficiency and acceleration. Two different parameters.

Where does acceleration come from? Could come from a variety of sources, including general maturation of memory and attention, specific changes in word recognition or processing (Fernald et al. 1998; **frank2026?**), the increasing sophistication of inferential strategies for word learning [(**smith2002?**); Markman and Wachtel (1988); etc], or other sources.

Mitchell and McMurray (2009) prove that acceleration does not result from simply leverage of vocabulary –

Role of input: Coffey 2026 variance decomposition

Furthermore, how does the accumulation process relate to the rapid growth of vocabulary? On some models, accumulation by itself can produce an exponential “vocabulary explosion” while the child’s own learning abilities remain constant. On other accounts, however, the child’s own abilities change over development. These changes could be due to the accumulation of language allowing bootstrapping (e.g., via mutual exclusivity or the shape bias), they could be through faster processing (Fernald rich get richer), or they could be via general maturation (e.g., better memory, faster processing). Or all of these.

Thus, modeling human learning requires positing other sources of variability and acceleration. Sources of variability could include from variation in interactional input quality, genetic variation between children, environmental richness not captured by input, etc. Sources of acceleration could include increases in processing efficiency, learning-to-learn including mutual exclusivity, shape bias, syntactic bootstrapping, as well as domain general changes in perception, processing speed, memory, attention or other aspects of cognition. The key to understanding children’s data efficiency may thus lie in better understanding when and how children’s learning accelerates with maturation and experience.

Continuous variation or clusters? (Pinstock and Schramm 2026; Singletary et al. 2025) We think it’s just continuous variation (Supplement)

Network models and their predictions e.g. (**steyvers2005?**; **hills2009?**; Fourtassi, Bian, and Frank 2020)

Methods

Data and Code

Data were retrieved from Wordbank using the **wordbankr** package (Frank et al. 2017), **chldes-db** using **chldesr** (Sanchez et al. 2019), and **peekbank** using **peekbankr** (**zettersten2022?**). BabyView data are from release 2025.2, specifically those used by (**tan2026?**). SEEDLings data are downloaded from the materials provided by Egan-Dailey and Bergelson (2025).

Reproducible code for this paper is available at <http://github.com/langcog/standard-model-2>.

Models

Frequentist models

Initial models were fit with `glmer` (`glmer?`), with the following specifications:

- M1: `produces ~ offset(log_age) + (1 | item)`
- M2: `produces ~ 1 + log_age + (1 | item)`
- M3: `produces ~ 1 + log_age + (1 | item) + (1 | child)`
- M4: `produces ~ 1 + log_age + (1 | item) + (log_age | child)`

Linear variants reported in the supplement used linear age rather than log age.

Bayesian models

Bayesian models were fit using Stan (`stan?`).

Priors

Inference

Input

GPT-2 models

We trained XYZ

References

Claude Code was used for literature review, code generation, visualization, and simulation infrastructure. All writing is original to the author.

- Adams, Katherine A., Virginia A. Marchman, Elizabeth C. Loi, Melanie D. Ashland, Anne Fernald, and Heidi M. Feldman. 2018. “Caregiver Talk and Medical Risk as Predictors of Language Outcomes in Full Term and Preterm Toddlers.” *Child Development* 89 (5): 1674–90.
- Anderson, Nina J., Susan A. Graham, Heather Prime, Jennifer M. Jenkins, and Sheri Madigan. 2021. “Linking Quality and Quantity of Parental Linguistic Input to Child Language Skills: A Meta-Analysis.” *Child Development* 92 (2): 484–501.
- Bloom, Paul. 2002. *How Children Learn the Meanings of Words*. MIT press.
- Bond, Trevor G, and Christine M Fox. 2013. *Applying the Rasch Model: Fundamental Measurement in the Human Sciences*. Psychology Press.
- Braginsky, Mika, Daniel Yurovsky, Virginia A Marchman, and Michael C Frank. 2019. “Consistency and Variability in Children’s Word Learning Across Languages.” *Open Mind* 3: 52–67.
- Chang, Tyler A., and Benjamin K. Bergen. 2022. “Word Acquisition in Neural Language Models.” *Transactions of the Association for Computational Linguistics* 10: 1–16. https://doi.org/10.1162/tacl_a_00444.
- Coffey, Joseph R., Okko Räsänen, Camila Scaff, and Alejandrina Cristia. 2024. “The Difficulty and Importance of Estimating the Lower and Upper Bounds of Infant Speech Exposure.” In *Proceedings of Interspeech 2024*, 3615. Kos, Greece. <https://doi.org/10.21437/Interspeech.2024-2102>.

- Coffey, Joseph R., and Jesse Snedeker. 2026. “How Strong Is the Relationship Between Caregiver Speech and Language Development? A Meta-Analysis.” *Journal of Child Language* 53: 435–70. <https://doi.org/10.1017/S0305000924000692>.
- De Boeck, Paul, Marjan Bakker, Robert Zwitser, Michel Nivard, Abe Hofman, Francis Tuerlinckx, and Ivailo Partchev. 2011. “The Estimation of Item Response Models with the Lmer Function from the Lme4 Package in r.” *Journal of Statistical Software* 39: 1–28.
- Egan-Dailey, Shannon, and Erika Bergelson. 2025. “Early Child Measures Outpredict Input Measures of Preschool Language Skills in US English Learners.” *Developmental Psychology*.
- Eriksson, Mårten, Peter B Marschik, Tiia Tulviste, Margareta Almgren, Miguel Pérez Pereira, Sonja Wehberg, Ljubica Marjanović-Umek, Frederique Gayraud, Melita Kovacevic, and Carlos Gallego. 2012. “Differences Between Girls and Boys in Emerging Language Skills: Evidence from 10 Language Communities.” *British Journal of Developmental Psychology* 30 (2): 326–43.
- Fenson, Larry et al. 2007. *MacArthur-Bates Communicative Development Inventories*. Baltimore, MD: Paul H. Brookes Publishing Company.
- Fenson, Larry, Philip S Dale, J Steven Reznick, Elizabeth Bates, Donna J Thal, Stephen J Pethick, Michael Tomasello, Carolyn B Mervis, and Joan Stiles. 1994. “Variability in Early Communicative Development.” *Monographs of the Society for Research in Child Development*, i–185.
- Fernald, Anne, and Virginia A Marchman. 2012. “Individual Differences in Lexical Processing at 18 Months Predict Vocabulary Growth in Typically Developing and Late-Talking Toddlers.” *Child Development* 83 (1): 203–22.
- Fernald, Anne, Virginia A Marchman, and Adriana Weisleder. 2013. “SES Differences in Language Processing Skill and Vocabulary Are Evident at 18 Months.” *Developmental Science* 16 (2): 234–48.
- Fernald, Anne, John P. Pinto, Daniel Swingley, Amy Weinberg, and Gerald W. McRoberts. 1998. “Rapid Gains in Speed of Verbal Processing by Infants in the 2nd Year.” *Psychological Science* 9 (3): 228–31.
- Fourtassi, Abdellah, Yuan Bian, and Michael C Frank. 2020. “The Growth of Children’s Semantic and Phonological Networks: Insight from 10 Languages.” *Cognitive Science* 44 (7): e12847. <https://doi.org/10.1111/cogs.12847>.
- Frank, Michael C. 2023. “Bridging the Data Gap Between Children and Large Language Models.” *Trends in Cognitive Sciences* 27 (11): 990–92.
- Frank, Michael C, Mika Braginsky, Daniel Yurovsky, and Virginia A Marchman. 2021. *Variability and Consistency in Early Language Learning: The Wordbank Project*. MIT Press.
- Frank, Michael C, Mika Braginsky, Daniel Yurovsky, and Virginia A. Marchman. 2017. “Wordbank: An Open Repository for Developmental Vocabulary Data.” *Journal of Child Language* 44 (3): 677–94.
- Ganger, Jennifer, and Michael R. Brent. 2004. “Reexamining the Vocabulary Spurt.” *Developmental Psychology* 40 (4): 621–32. <https://doi.org/10.1037/0012-1649.40.4.621>.
- Gómez Díaz, Miranda, Laia Fibla, Rachel Ka-Ying Tsui, and Krista Byers-Heinlein. 2024. “Testing Theories of the Vocabulary Spurt with Monolingual and Bilingual Infants.” *Developmental Psychology* 60 (8): 1357–71. <https://doi.org/10.1037/dev0001777>.
- Goodman, Judith C, Philip S Dale, and Ping Li. 2008. “Does Frequency Count? Parental Input and the Acquisition of Vocabulary.” *Journal of Child Language* 35 (3): 515–31.
- Grimm, Kevin J, Nilam Ram, and Ryne Estabrook. 2016. *Growth Modeling: Structural Equation*

- and *Multilevel Modeling Approaches*. Guilford Publications.
- Hagihara, Hiromichi, Monica Barbir, Mikako Ishibashi, Yasuhiro Kanakogi, Masaharu Kato, Irena Lovcevic, Youtao Lu, et al. 2023. “A Sharable Merged Dataset on Japanese Children’s Vocabulary Measures Using the Japanese MacArthur–Bates Communicative Development Inventory.” OSF. <https://doi.org/10.17605/osf.io/s5ydw>.
- Hart, Betty, and Todd R Risley. 2003. “The Early Catastrophe: The 30 Million Word Gap by Age 3.” *American Educator* 27 (1): 4–9.
- Hidaka, Shohei. 2013. “A Computational Model Associating Learning Process, Word Attributes, and Age of Acquisition.” *PloS One* 8 (11).
- Kachergis, George, Virginia A Marchman, Philip S Dale, Jessica Mankewitz, and Michael C Frank. 2022. “Online Computerized Adaptive Tests of Children’s Vocabulary Development in English and Mexican Spanish.” *Journal of Speech, Language, and Hearing Research* 65 (6): 2288–308.
- Kachergis, George, Virginia A Marchman, and Michael C Frank. 2022. “Toward a ‘Standard Model’ of Early Language Learning.” *Current Directions in Psychological Science* 31 (1): 20–27.
- Long, Bria, Robert Z Sparks, Violet Xiang, Stefan Stojanov, Zi Yin, Grace E Keene, Alvin WM Tan, et al. 2024. “The BabyView Dataset: High-Resolution Egocentric Videos of Infants’ and Young Children’s Everyday Experiences.” *arXiv Preprint arXiv:2406.10447*.
- MacWhinney, Brian. 2000. *The CHILDES Project: Tools for Analyzing Talk*. Vol. 1. Psychology Press.
- Marchman, Virginia A, Philip Dale, and Larry Fenson. 2021. *MacArthur-Bates Communicative Development Inventories User’s Guide and Technical Manual, Third Edition*. Brookes Publishing.
- Marchman, Virginia A, and Anne Fernald. 2008. “Speed of Word Recognition and Vocabulary Knowledge in Infancy Predict Cognitive and Language Outcomes in Later Childhood.” *Developmental Science* 11 (3): F9–16.
- Markman, Ellen M., and Gwyn F. Wachtel. 1988. “Children’s Use of Mutual Exclusivity to Constrain the Meanings of Words.” *Cognitive Psychology* 20 (2): 121–57.
- Markson, Lori, and Paul Bloom. 1997. “Evidence Against a Dedicated System for Word Learning in Children.” *Nature* 385 (6619): 813–15.
- Mayor, Julien, and Kim Plunkett. 2010. “Vocabulary Spurt: Are Infants Full of Zipf?” In *Proceedings of the 32nd Annual Conference of the Cognitive Science Society*. Vol. 32.
- McMurray, Bob. 2007. “Defusing the Childhood Vocabulary Explosion.” *Science* 317 (5838): 631–31.
- Mitchell, C., and B. McMurray. 2009. “On Leveraged Learning in Lexical Acquisition and Its Relationship to Acceleration.” *Cognitive Science* 33 (8): 1503–23.
- Mollica, Francis, and Steven T Piantadosi. 2017. “How Data Drive Early Word Learning: A Cross-Linguistic Waiting Time Analysis.” *Open Mind* 1 (2): 67–77.
- Pinstock, Eveline, and Satyam Antonio Schramm. 2026. “Vocabulary Trajectories in German-Speaking Children from 18 Months to Three Years: A Growth Mixture Model.” *Journal of Child Language* 53 (1): 62–79. <https://doi.org/10.1017/S0305000924000503>.
- Rasch, Georg. 1960. *Probabilistic Models for Some Intelligence and Attainment Tests*. Copenhagen, Denmark: Danish Institute for Educational Research.
- Rescorla, Leslie. 2011. “Late Talkers: Do Good Predictors of Outcome Exist?” *Developmental Disabilities Research Reviews* 17 (2): 141–50.
- Roy, Brandon C., Michael C Frank, Philip DeCamp, Matthew Miller, and Deb Roy. 2015. “Predicting the Birth of a Spoken Word.” *Proceedings of the National Academy of Sciences* 112 (41): 12663–68.

- Sanchez, Alessandro, Stephan C. Meylan, Mika Braginsky, Kyle E. MacDonald, Daniel Yurovsky, and Michael C Frank. 2019. “Childes-Db: A Flexible and Reproducible Interface to the Child Language Data Exchange System.” *Behavior Research Methods* 51 (4): 1928–41.
- Simonsen, Hanne Gram, Kristian Emil Kristoffersen, Dorthé Bleses, Sonja Wehberg, and Rune Nørgård Jørgensen. 2014. “The Norwegian Communicative Development Inventories: Reliability, Main Developmental Trends and Gender Differences.” *First Language* 34 (1): 3–23. <https://doi.org/10.1177/0142723713510997>.
- Singletary, Britt, Hui Jiang, Winifred Graham Wilberforce, Daniela Avelar, Kristina Strother-Garcia, and Laura M. Justice. 2025. “Examining Early Vocabulary Growth Trajectories in Late Talkers in a Low-Income Longitudinal Sample.” *Infancy* 30 (4): e70036. <https://doi.org/10.1111/inf.70036>.
- Thal, Donna J., Virginia A. Marchman, and J. Bruce Tomblin. 2013. “Late Talking Toddlers: Characterization and Prediction of Continued Delay.” In *Late Talkers: Language Development, Interventions, and Outcomes*, edited by L. Rescorla and P. Dale. Baltimore, MD: Brookes Publishing.
- Trueswell, John C, Tamara Nicol Medina, Alon Hafri, and Lila R Gleitman. 2013. “Propose but Verify: Fast Mapping Meets Cross-Situational Word Learning.” *Cognitive Psychology* 66 (1): 126–56.
- Warstadt, Alex, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, et al. 2023. “Findings of the BabyLM Challenge: Sample-Efficient Pretraining on Developmentally Plausible Corpora.” In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, 1–34.
- Weisleder, A., and A. Fernald. 2013. “Talking to Children Matters: Early Language Experience Strengthens Processing and Builds Vocabulary.” *Psychological Science* 24 (11): 2143–52. <https://doi.org/10.1177/0956797613488145>.
- Woodward, Amanda L, Ellen M Markman, and Colleen M Fitzsimmons. 1994. “Rapid Word Learning in 13-and 18-Month-Olds.” *Developmental Psychology* 30 (4): 553.